

EX:01**Design and Evaluate Different Prompt Types to Control Agent Reasoning, Autonomy, and Output Quality****Aim:**

To design and evaluate different prompt engineering techniques for controlling an AI agent's reasoning process, autonomy, and output quality, and compare their effectiveness on the same task.

Procedure:

1. Select a common task (e.g., text summarization, question answering, or explanation).
2. Design different prompt types:
 - o Zero-Shot Prompt
 - o Few-Shot Prompt
 - o Role-Based Prompt
 - o Chain-of-Thought Prompt
3. Provide the same input to the AI model using each prompt.
4. Collect the generated responses.
5. Evaluate each response based on:
 - o Accuracy
 - o Reasoning Quality
 - o Completeness
 - o Output Format
 - o Response Length
6. Compare the outputs and record observations.
7. Conclude which prompt type performs best for the selected task.

Code:

```
from google import genai

client = genai.Client(api_key="YOUR_GEMINI_API_KEY")

task = "Explain why cybersecurity is important for online banking."

prompts = {
    "Zero-Shot":
        task,

    "Few-Shot":
        """
```

Example:

Question: What is Machine Learning?

Answer: Machine Learning enables computers to learn patterns from data and improve their performance without being explicitly programmed.

Now answer:

Explain why cybersecurity is important for online banking.

```

"""
    "Role-Based":
        """
You are a banking security analyst.
Explain why cybersecurity is important for online banking.
"""
    "Chain-of-Thought":
        """
Think step by step before answering.

Question:
Explain why cybersecurity is important for online banking.
"""
}

```

```

# Generate responses for each prompt
for prompt_type, prompt in prompts.items():

```

```

    response = client.models.generate_content(
        model="gemini-2.5-flash",
        contents=prompt
    )

```

```

    print("=" * 60)
    print(prompt_type)
    print("=" * 60)
    print(response.text)

```

```

    print()

```

Output:

```

=====
Zero-Shot
=====

```

Cybersecurity is essential for online banking because banks handle highly sensitive financial data and money, which makes them a constant target for criminals. Here's a breakdown of why it matters:

1. **Protecting Customer Funds and Accounts:**

Preventing Unauthorized Access: Attackers use stolen passwords, phishing emails and malware to take over accounts and transfer money. Strong authentication and monitoring help stop these attempts before funds are lost.

Securing Transactions: Encryption protects payment details while they travel between the customer's device and the bank's servers, so they cannot be intercepted or altered.

2. **Safeguarding Personal and Financial Data:**

Identity Theft Prevention: Online banking systems store names, addresses, account numbers and transaction histories. If this data leaks, it can be used for identity theft, loan fraud and impersonation.

* **Regulatory Compliance:** Banks must follow data protection and security regulations. Weak security can lead to heavy fines and legal action.

3. **Maintaining Trust and Reputation:**

* **Customer Confidence:** Customers will only use digital banking if they believe their money is safe. A single breach can cause people to leave and damage a bank's reputation for years.

4. **Keeping Services Available:**

* **Defense Against Attacks:** Denial-of-service attacks and ransomware can take banking platforms offline. Good cybersecurity keeps services running and limits the damage when attacks occur.

5. **Countering Evolving Threats:**

* **Constant Adaptation:** Attack methods such as social engineering, fake banking apps and SIM-swap fraud keep changing, so banks need continuous monitoring, updates and staff training.

In essence, cybersecurity protects money, personal data, and trust, which are the foundations of online banking.

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Few-Shot

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Answer: Cybersecurity is important for online banking because it protects customers' money and personal data from fraud, hacking, and identity theft.

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Role-Based

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As a banking security analyst, I can say that cybersecurity is not an optional add-on for online banking. It is the **foundation** that makes digital banking possible. The key reasons are:

1. **Defending Against Fraud and Account Takeover:**

* **The Threat:** Credential stuffing, phishing, vishing and malware-based attacks target customers every day. Once an attacker controls an account, funds can be moved out within minutes through mule accounts.

* **The Control:** Multi-factor authentication, device fingerprinting, behavioural analytics and real-time fraud scoring detect unusual logins or transfers and block them before settlement.

2. **Protecting Data in Transit and at Rest:**

* **The Threat:** Man-in-the-middle attacks on insecure networks and breaches of databases can expose card numbers, account details and personal information.

* **The Control:** TLS encryption, tokenization of card data, strong key management and strict access controls ensure stolen data is useless to an attacker.

3. **Securing APIs and Third-Party Integrations:**

* **The Threat:** Open banking APIs, mobile apps and fintech partners widen the attack surface. A weakness in one integration can expose the whole system.

* **The Control:** API gateways, rate limiting, OAuth-based consent, regular penetration testing and vendor risk assessments reduce this exposure.

4. **Meeting Regulatory and Audit Requirements:**

* **The Obligation:** Banks operate under strict regulations such as PCI DSS, central bank cyber-resilience frameworks and data protection laws. Failure to comply leads to penalties and loss of operating licences.

* **The Benefit:** A mature security programme with logging, incident response plans and audits shows regulators and customers that risk is managed.

5. **Ensuring Resilience and Business Continuity:**

* **The Threat:** Ransomware and distributed denial-of-service attacks can disrupt payments, salaries and everyday transactions for thousands of customers.

* **The Control:** Network segmentation, backups, failover systems and tested incident response plans allow a quick recovery with minimal customer impact.

6. **Preserving Customer Trust:**

* **The Impact:** Trust is the real currency of banking. Customers who feel unsafe move to competitors, and the reputational damage from a breach often costs more than the direct financial loss.

In conclusion, strong cybersecurity protects money, data and continuity of service, and it is the reason customers can safely rely on online banking every day.

=====
Chain-of-Thought
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Online banking is important to billions of people, and cybersecurity is important because it protects the people, money and systems involved. Here is the reasoning step by step:

1. **Identify What Is at Stake:**

* **Money and Data:** Online banking systems hold account balances, transaction histories, card details and personal identity information. All of these have direct financial value to criminals.

2. **Identify Who Might Attack and How:**

* **Common Threats:** Phishing, malware, fake banking apps, SIM-swap fraud, insider misuse and attacks on the bank's servers are all realistic threats because banking is accessed over the open internet.

3. **Consider the Consequences of an Attack:**

* **For Customers:** Loss of savings, damaged credit, stress and the long process of recovering a stolen identity.

* **For Banks:** Financial losses, regulatory penalties, legal costs and loss of customer trust.

* **For the Economy:** Large-scale incidents can disrupt payments and reduce confidence in the financial system.

4. **Determine What Protection Is Needed:**

* **Prevention:** Encryption, multi-factor authentication, secure coding and regular patching make attacks harder.

* **Detection and Response:** Fraud monitoring, alerts and incident response plans limit the damage when something goes wrong.

* **Awareness:** Educating customers and staff reduces the success of social engineering.

5. **Reach a Conclusion:**

* Since online banking combines high-value assets, constant exposure to the internet and serious consequences for failure, strong cybersecurity is necessary for keeping money, data and trust safe.

In summary, cybersecurity is not just a technical feature of online banking. It is a basic requirement for a safe, reliable and trusted financial system.

Evaluation:

...	Prompt Type	Accuracy	Relevance	Completeness	Clarity	Reasoning \
0	Zero-Shot	5	5	5	4	4
1	Few-Shot	5	4	3	5	3
2	Role-Based	5	5	5	5	5
3	Chain-of-Thought	5	5	5	4	5
	Total	Output Quality (%)				
0	23	92.0				
1	20	80.0				
2	25	100.0				
3	24	96.0				
Evaluation completed successfully.						

Result:

The Gemini API successfully generated responses for each prompt type. The experiment shows that:

- **Zero-Shot Prompting** gives a clear and well-structured answer, but it follows a generic explanation style.
- **Few-Shot Prompting** copies the length and style of the example, so the answer is very short and less complete.
- **Role-Based Prompting** produces the most detailed and practical answer, with threats and controls explained from a security analyst's view.
- **Chain-of-Thought Prompting** presents the answer as a logical sequence of steps, which makes the reasoning easy to follow.

Overall, Role-Based Prompting achieved the highest output quality (100%) for this task, followed by Chain-of-Thought Prompting (96%), Zero-Shot Prompting (92%) and Few-Shot Prompting (80%).